Bayesian meta-analysis in Stan with *baggr*

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baggr

baggr (Bayesian aggregator) is a new R package for Bayesian meta-analysis models that uses Stan

Our goals:

1. Implement all basic meta-analysis models and tools
2. Focus on accessibility, model criticism and comparison
3. Help people avoid basic mistakes
4. Keep the framework flexible and extend to more models

(Probably) *not* our goal:

5. Build a package for people who already build their models in Stan

baggr beta is available on CRAN and GitHub

Basic model of meta-analysis is simple...

After Rubin (1981)[1]

- Assume only aggregate data (y, se) are known ($y_i \sim \mathcal{N}(\tau_i, se_i)$)
  - We treat se_i as fixed (works OK with big samples)
- Hierarchical model: $\tau_i \sim \mathcal{N}(\tau, \sigma)$
  - $\sigma = 0$ is "full pooling", $\sigma = \infty$ is "no pooling"
  - Fixed and random effects in medical stats
  - Typically it's more sensible to start with $0 < \sigma < \infty$
  - Note how the amount of heterogeneity across i's will be controlled by σ prior

[1] Rubin, Donald B. "Estimation in Parallel Randomized Experiments". Journal of Educational Statistics 6, no. 4 (1981): 377–401.

. . . but it gets more complicated fast

- Continuous, categorical data
- Meta-analysis of means (summary data)
- Meta-regression
- Individual-level data (or a mix of summary and individual)
- Modeling more parameters (mixture data, variances, quantiles)
- Dealing with small samples (e.g. chemical risk assessment)
- Model-based meta-analysis (drug development)
- Network meta-analysis / ITC (health technology assessment)

Techniques, terminology, *motivations* and even software are very domain-specific

Example (8 schools!)

We use a familiar summary-level dataset from Rubin (1981).
Individual-level data works just as well.

```
library(baggr)
schools
```

```
##      group tau se
## 1 School A  28 15
## 2 School B   8 10
## 3 School C  -3 16
## 4 School D   7 11
## 5 School E  -1  9
## 6 School F   1 11
## 7 School G  18 10
## 8 School H  12 18
```

baggr syntax example: simple model workflow

```
# Fitting function is baggr()
fit <- baggr(schools, model = "rubin", pooling = "partial",
             prior_hypermean = normal(0, 25), prior_hypersd = uniform(0, 100))

# Semi-automatic prior choice, automatic model choice (with prompts):
eff_lab <- "mean SAT improvement"
fit_c <- baggr(schools, prior_hypersd = cauchy(0, 50), effect = eff_lab)
fit_n <- baggr(schools, prior_hypersd = normal(0, 10), effect = eff_lab)
fit_u <- baggr(schools, prior_hypersd = uniform(0, 100),
                        prior_hypermean = normal(-5, 10), effect = eff_lab)

# Plot results for one model, making use of bayesplot
baggr_theme_update(text = element_text(family = "mono"))
plot(fit_c, style = "areas", hyper = TRUE, order = TRUE)

# Compare fit to data
bgc <- baggr_compare("Cauchy" = fit_c, "Normal" = fit_n, "Uniform" = fit_u)
bgc; plot(bgc)
# Compare posterior predictive distribution (note ggplot)
effect_plot("Cauchy" = fit_c, "Normal" = fit_n, "Uniform" = fit_u) + xlim(-10, 30)

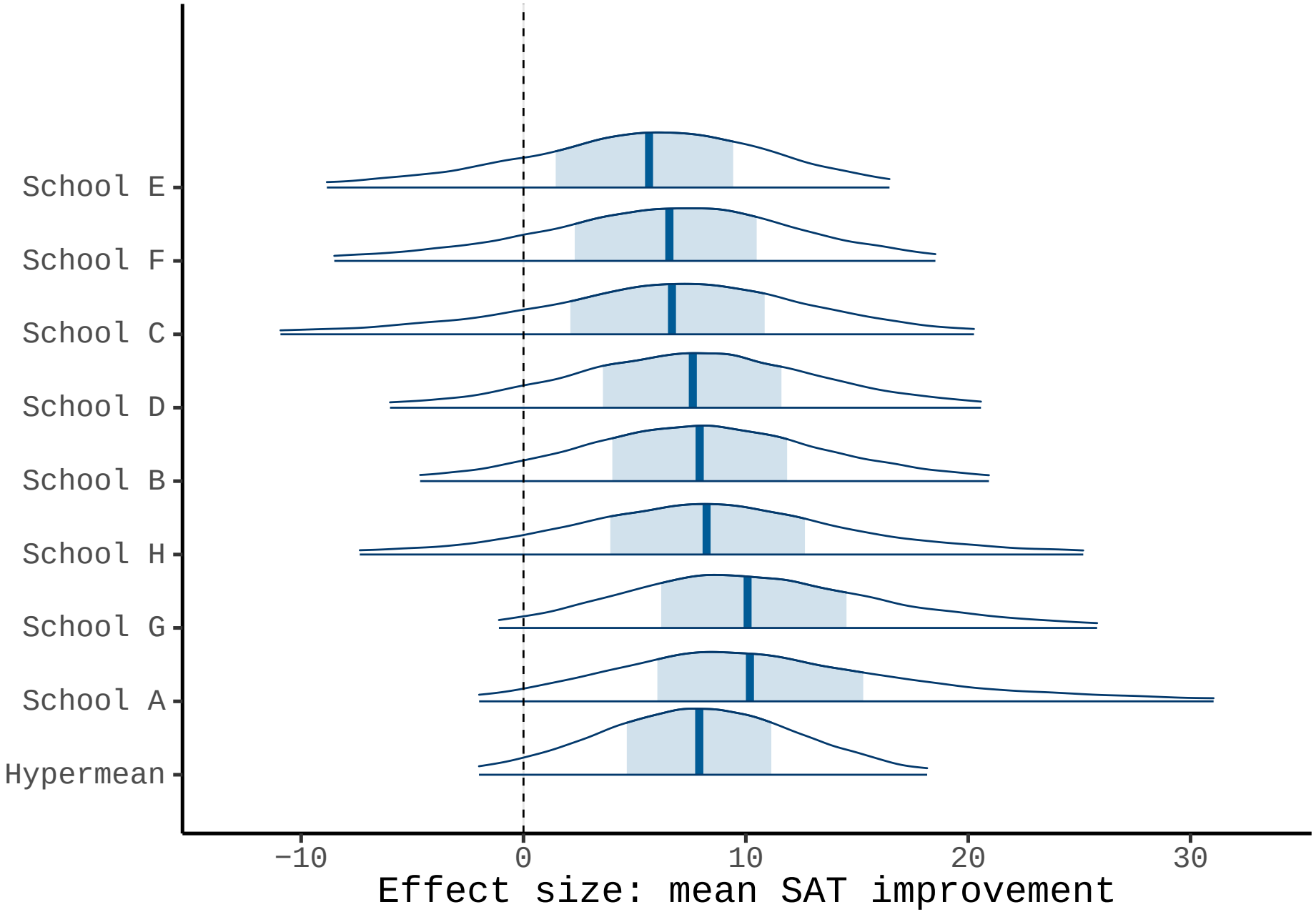
# Compare the impact of different prior choices
bgc_pooling <- baggr_compare(schools, prior_hypersd = normal(0, 10), what = "pooling")
plot(bgc_pooling)

# Use LOO cross-validation to choose between partial & full pooling
loo_p <- loocv(schools, pooling = "partial")
loo_f <- loocv(schools, pooling = "full")
loo_compare(loo_p, loo_f)
```

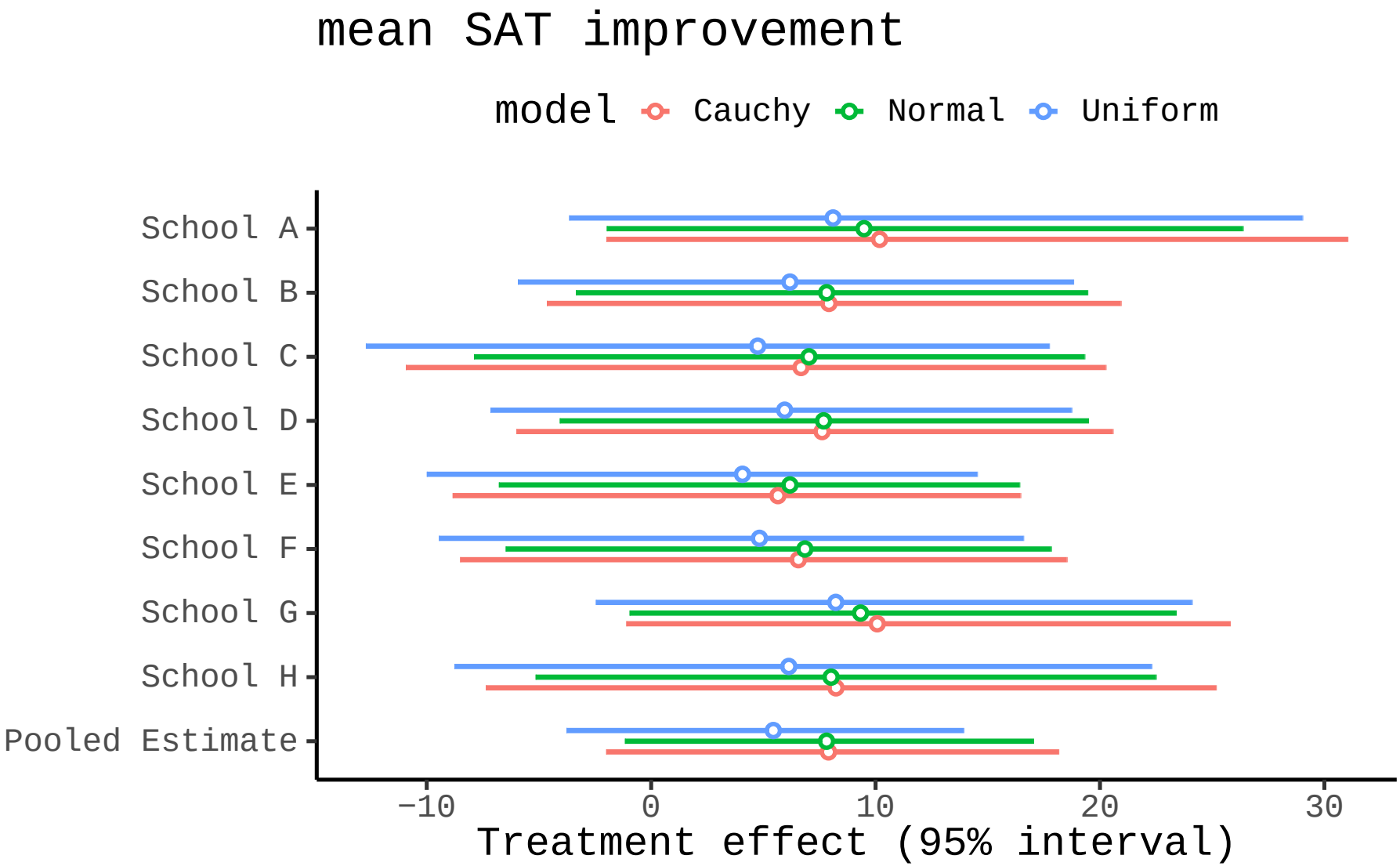
```
fit_c
```

```
## Model type: Rubin model with aggregate data
## Pooling of effects: partial
##
## Aggregate treatment effect (on mean SAT improvement):
## Hypermean (tau) =  7.9 with 95% interval -2.0 to 18.2
## Hyper-SD (sigma_tau) = 6.39 with 95% interval 0.25 to 19.81
## Total pooling (1 - I^2) = 0.8 with 95% interval 0.3 to 1.0
##
## Treatment effects on mean SAT improvement:
##          mean  sd pooling
## School A 11.3 8.1    0.83
## School B  8.0 6.3    0.73
## School C  6.2 7.6    0.84
## School D  7.6 6.5    0.76
## School E  5.2 6.3    0.70
## School F  6.2 6.7    0.76
## School G 10.7 6.8    0.73
## School H  8.4 7.8    0.87
```

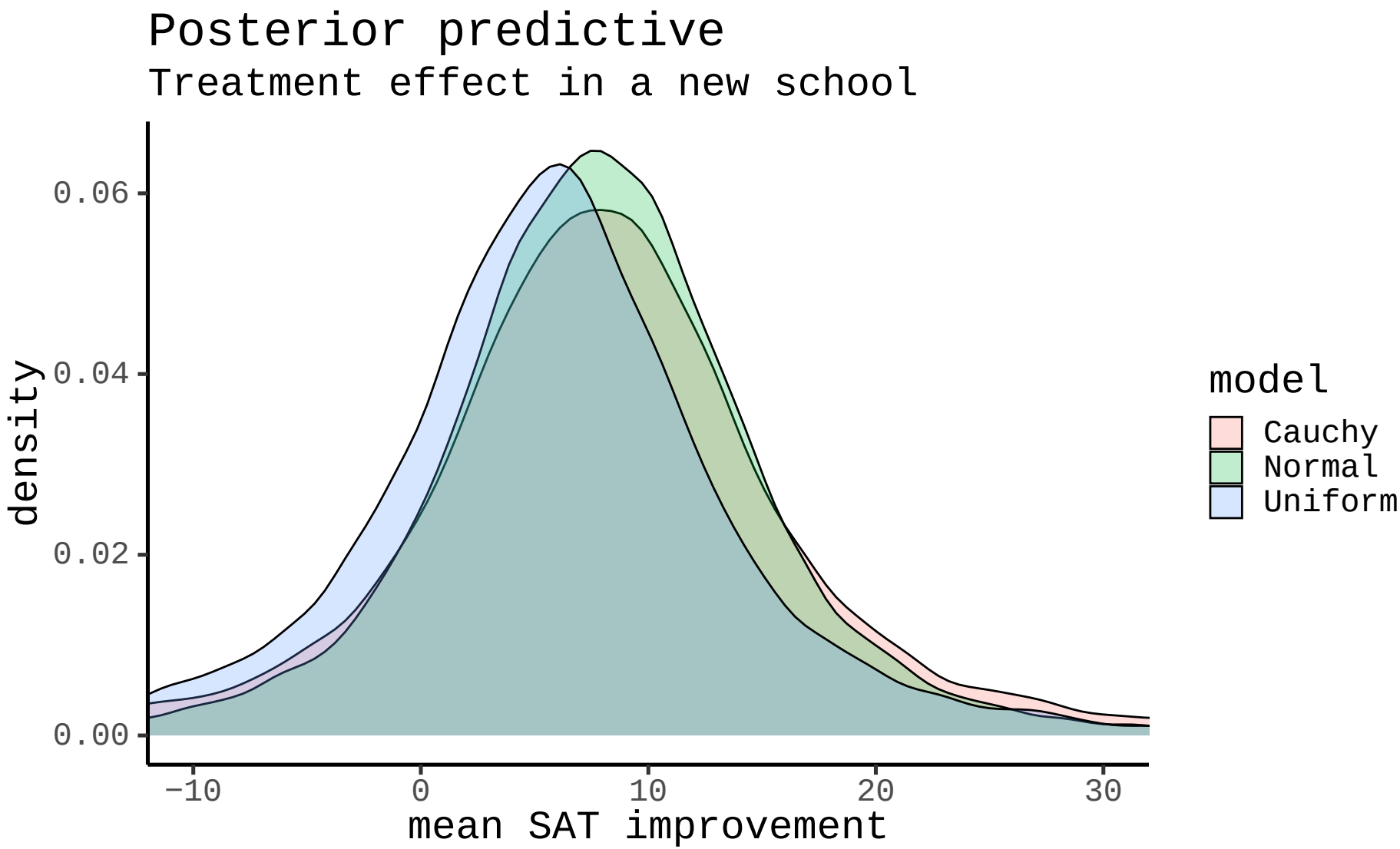
```
plot(fit_c, style = "areas", hyper = TRUE, order = TRUE)
```



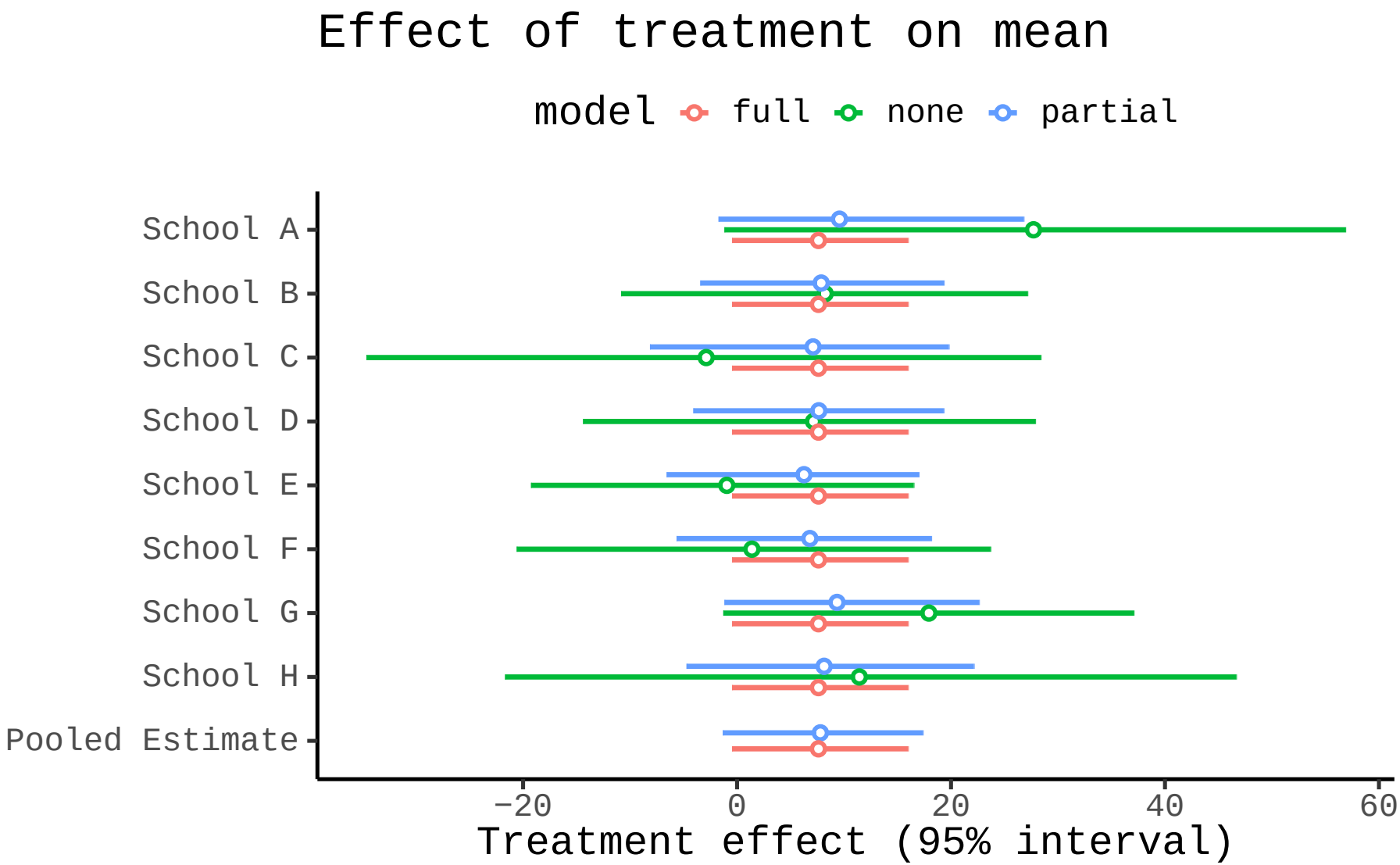
```
bgc <- baggr_compare("Cauchy" = fit_c, "Normal" = fit_n, "Uniform" = fit_u)
bgc; plot(bgc)
```



```r
effect_plot("Cauchy" = fit_c, "Normal" = fit_n, "Uniform" = fit_u) +
  coord_cartesian(xlim = c(-10, 30)) +
  ggtitle("Posterior predictive", "Treatment effect in a new school")
```



```r
bgc_pooling <- baggr_compare(schools, prior_hypersd = normal(0, 10),
                             what = "pooling")
plot(bgc_pooling)
```



```
loo_p <- loocv(schools, pooling = "partial")
loo_f <- loocv(schools, pooling = "full")
loo_compare(loo_p, loo_f)
```

```
## Comparison of cross-validation
##
##                       ELPD ELPD SE
## Model 1 - Model 2 -0.731   0.343
```

Additional commands:

```
treatment_effect(fit)
group_effects(fit)
effect_draw(fit)
forest_plot(fit)
pooling(fit)
```

Going beyond the simple use cases

Already included in the new release (v0.7, July 2020):

- Models with many parameters (quantiles, mixtures, effects on means & variances)
- Running models on individual-level data
- Including covariates (meta-regression or individual-level)
- Working with binary data (rare events, L'Abbe and bubble plots)
- Extra summaries, transforms and plots of data
- Tutorial vignettes

How to make the tool better? Our 5 questions

How to...

- Improve syntax and/or make it more flexible
- Reach more users
- Understand who does and who doesn't need help
- Give more technical users means of extending their models
- Connect more to Stan ecosystem (e.g. `bayesplot`)

Each of these areas is where feedback from Stan community is especially valuable

Thank you

baggr on GitHub: `github.com/wwiecek/baggr/`

Find a copy of slides on `wwiecek.github.io`